

LiftLab - Gym Management Dashboard

1. PROJECT DETAILS

Title of the Project:

LIFTLAB - Gym Management Dashboard

Purpose of the Project:

LiftLab is a premium, full-featured gym management dashboard built with Next.js 16, React, and a liquid-glass design system. It serves as a comprehensive command center for gym owners and administrators to manage members, staff, class schedules, machines, nutrition plans, supplements, fees, and generate detailed reports. The platform implements four core data structures (Stack, Queue, QuickSort, Hash Map) as required by the problem statement, and features a six-color palette design system with light and dark themes. The goal is to provide real-time floor intelligence, member progress tracking, and maintenance signals in a single, visually polished interface.

Target Users:

- **Gym Owners & Administrators:** Decision-makers who need a real-time overview of operations, revenue, member retention, and equipment status.
- **Personal Trainers:** Staff who manage member workout plans, track progress, and need quick access to member profiles and HR zone safety tools.
- **Nutritionists:** Professionals who create and assign meal plans, track macro compliance, and manage supplement stacks for members.
- **Front Desk Staff:** Personnel who handle member onboarding, fee collection, class enrollment, and daily check-ins.
- **ITM Skills University Evaluators:** Academic reviewers assessing the implementation of data structures, React patterns, and design system principles.

Tools Used:

- **Next.js 16 (App Router):** React framework providing SSR, file-based routing, and Turbopack for fast development.

LiftLab - Gym Management Dashboard

- **jsPDF 4:** Client-side PDF generation for fee receipts, staff payslips, and five report types.
- **next-themes:** Light/dark theme switching with no flash of unstyled content (FOUC).
- **Sora, Inter, JetBrains Mono:** Google Fonts for display headings, body text, and tabular number data.

2. KEY FEATURES

- **Liquid Glass Design System:** Translucent layered surfaces with backdrop-filter blur, specular edge highlights, and a strict six-color palette (#F6F6F6, #E0DBF3, #F3BA60, #B6B1C0, #736A6A, #202022) used across both light and dark themes.
- **Nine Functional Modules:** Dashboard, Members (list + 8-tab profile), Staff (list + 5-tab profile), Schedule, Machines, Nutrition, Supplements, Fees, and Reports — all navigable via a Zustand-driven SPA router.
- **Four DSA Implementations:** Stack-based workout undo/redo (useReducer), FIFO sensor data queue (useRef + setInterval), in-place QuickSort for progress ranking, and Hash Map O(1) trainer ID verification.
- **Dynamic State Management:** Members CRUD via Zustand store — adding a member updates the list instantly. Notifications persist dismissed state to localStorage so they never reappear.
- **Working Add Member Modal:** Full form with validation, auto-generated member ID and photo, starter workout plan, and toast confirmation.
- **Settings Modal:** Theme toggle, compact mode, dashboard widget toggles, currency selection — all persisted across reloads.
- **Safe Training Planner:** Heart rate zone calculator (5 zones, Max HR = 220 minus age) with medical-aware safety checks that alert when Zone 5 time exceeds 20 minutes for flagged members.

LiftLab - Gym Management Dashboard

3. COMPONENT DEFINITIONS USED IN THE PROJECT

3.1 Layout Components

Component	Definition
Sidebar.jsx	Dark #202022 left navigation with 9 module links, dynamic alert badges (expired members, machine alerts, overdue fees), active-state animated indicator, and a settings button.
Topbar.jsx	Dark sticky header with page title, Add Member button, search pill, clean HH:MM clock with seconds, theme toggle, notifications bell with dismissible dropdown, and profile avatar.
page.jsx	Root SPA shell that switches between nine modules based on Zustand navStore state. Includes ambient gradient backdrop and global modal rendering. No footer.

3.2 UI Primitives (src/components/ll/)

Component	Definition
Logo.jsx	Inline SVG barbell logo where the two weight plates form the letters L and L (LiftLab initials) in orange #F3BA60 with a white connecting bar.
GlassCard.jsx	Liquid glass container using backdrop-filter blur(24px) and saturate(160%) for translucent panels, modals, and overlays.
NeumorphCard.jsx	Neumorphic container with dual-shadow (dark plus light) for KPI stat cards and summary rows. Accepts a style prop for palette color rotation.
ClayCard.jsx	Claymorphism container with chunky 22px border radius and drop shadow for product cards and class schedule chips.
StatKPI.jsx	Animated KPI card using JetBrains Mono with tabular-nums for numbers. Supports cardColor prop to rotate through the six-color palette, auto-adapting text color (dark on light, light on dark).
StatusBadge.jsx	Color-coded pill badge that maps status strings to palette colors: orange for positive, brown for negative, purple-gray for neutral.
MachineIcon.jsx	Eight skeuomorphic SVG silhouettes (treadmill, elliptical, rowing, cable, squat-rack, bench, leg-press, smith) drawn with thin strokes. Includes shapeForMachine mapper and statusColor helper.
ThemeToggle.jsx	Animated moon-to-sun toggle using Framer Motion AnimatePresence with rotate and slide transitions. Calls next-themes setTheme.

3.3 State Management Stores (src/store/)

Store	Definition
navStore.js	Zustand store tracking the active module (dashboard, members, staff, etc.) and selected profile IDs for drill-down navigation. Not persisted.

LiftLab - Gym Management Dashboard

Store	Definition
appStore.js	Zustand store with persist middleware (skipHydration for SSR safety) managing members CRUD, dismissible notifications, and user settings. Only notifications and settings persist to localStorage; members re-seed each load.
uiStore.js	Zustand store controlling which global modal (Add Member, Settings) is open. Rendered at the root page level.

3.4 Custom Hooks (src/hooks/)

Hook	Definition
useAgeAutoUpdate.js	Exports BUSINESS_TODAY (pinned to 2025-02-15 for deterministic demo behavior), calcAge, membershipStatus, daysUntil, and machineServiceStatus. All date comparisons use BUSINESS_TODAY to prevent hydration mismatches.
useSensorQueue.js	FIFO queue hook generating sensor readings every 3 seconds. Uses useRef for the queue array (mutated in place) and useState for rendered list. Caps at 12 entries, auto-dequeuing the oldest.
useWorkoutHistory.js	Undo/redo stack implemented with useReducer. Maintains history and redo arrays. Push clears redo (standard branching undo). Exposes canUndo, canRedo, undo, redo, and reset.
useMaintenanceAlert.js	Memoized hook filtering machines by daysUntil(next_service_due). Returns overdueCount, dueSoonCount (within 7 days), alertMachines sorted by due date, and hasAlerts boolean.

3.5 DSA Utilities (src/utlis/)

Utility	Definition
progressSorter.js	In-place QuickSort (partition-based, $O(n \log n)$) sorting members by goal completion percentage. Also exports getMemberProgress (average of goal completion) and progressColor returning discrete palette colors (#736a6a, #b6b1c0, #f3ba60).
heartRateZone.js	Defines 5 HR zones with palette-only colors (b6b1c0, e0dbf3, f3ba60, 736a6a, 202022). Exports maxHR (220 minus age), zoneForHR, and isSafePlan which alerts when medical-flagged members exceed 20 minutes in Zone 5.
spaceBalancer.js	Gym zone occupancy system with 5 zones (Cardio 30, Weights 25, Machines 20, Group 40, Recovery 10). occupancyColor returns discrete palette colors. balanceStatus returns Balanced, Crowded, or Critical based on capacity thresholds.
format.js	Number formatting: inr() for Indian currency with lakh grouping (482000 becomes Rs.4,82,000), inrCompact() for L/Cr suffixes, num() for grouped numbers, pct() for percentages, pad() for zero-padded IDs.

LiftLab - Gym Management Dashboard

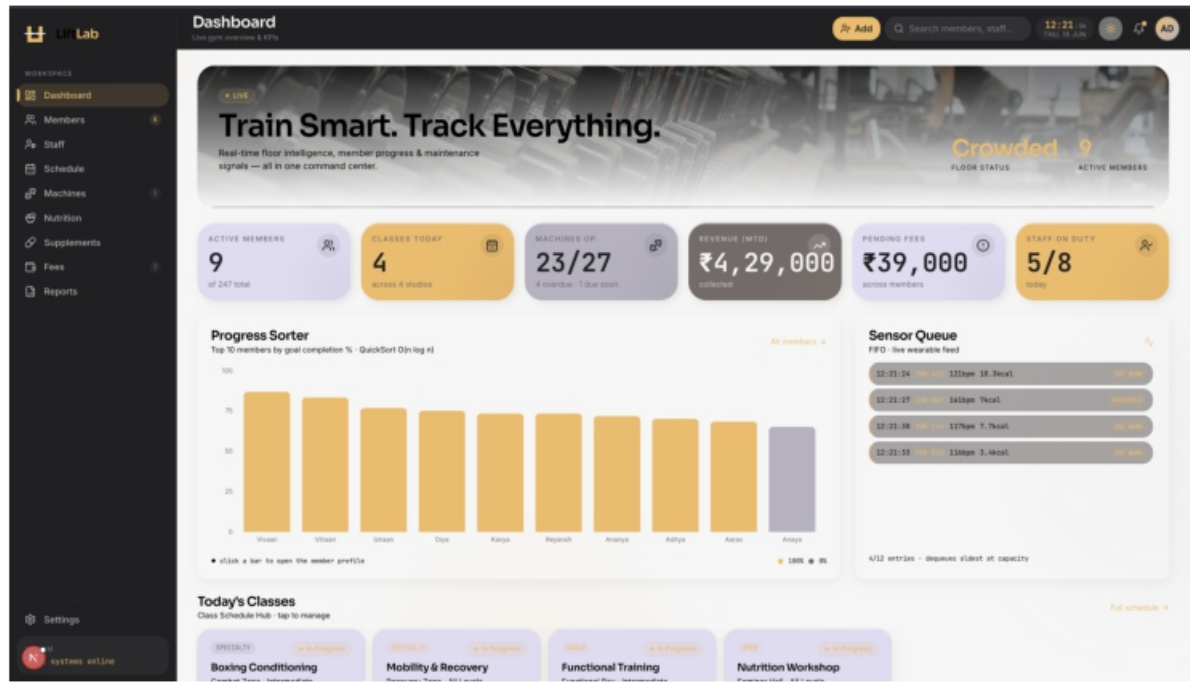
4. DESIGN DESCRIPTION

LiftLab uses a strict six-color palette design system inspired by modern minimalist aesthetics. The background is off-white #F6F6F6, with the sidebar, top bar using dark lead #202022. Cards rotate through four palette colors: lavender #E0DBF3, orange #F3BA60, purple-gray #B6B1C0, and brown #736A6A. All text is black. The design uses Sora as its display heading font, Inter for body text, and JetBrains Mono with tabular-nums for all numeric data, ensuring equal-width digits that do not jitter when values update live.

The project implements four distinct morphism styles applied contextually rather than uniformly. Glassmorphism (backdrop-filter blur 24-32px with saturate) is used for floating modal overlays, the top navigation bar, and sensor data panels. Neumorphism (dual shadow with inset highlight) is used for KPI stat cards and summary rows. Claymorphism (chunky 22px border radius with drop shadow) is used for supplement product cards and class schedule chips. Skeuomorphism is used for machine icons rendered as minimal SVG silhouettes of real gym equipment with status LED dots.

The application uses a single-page application architecture where a Zustand navStore controls which of the nine modules renders, avoiding Next.js route transitions for smoother UX. State is split into three stores: navStore for active module tracking, appStore for members CRUD and persisted notifications/settings, and uiStore for modal state. The appStore uses Zustand persist middleware with skipHydration to prevent SSR re-render loops, and only persists notifications and settings to localStorage while members re-seed from data files on each load.

LiftLab - Gym Management Dashboard

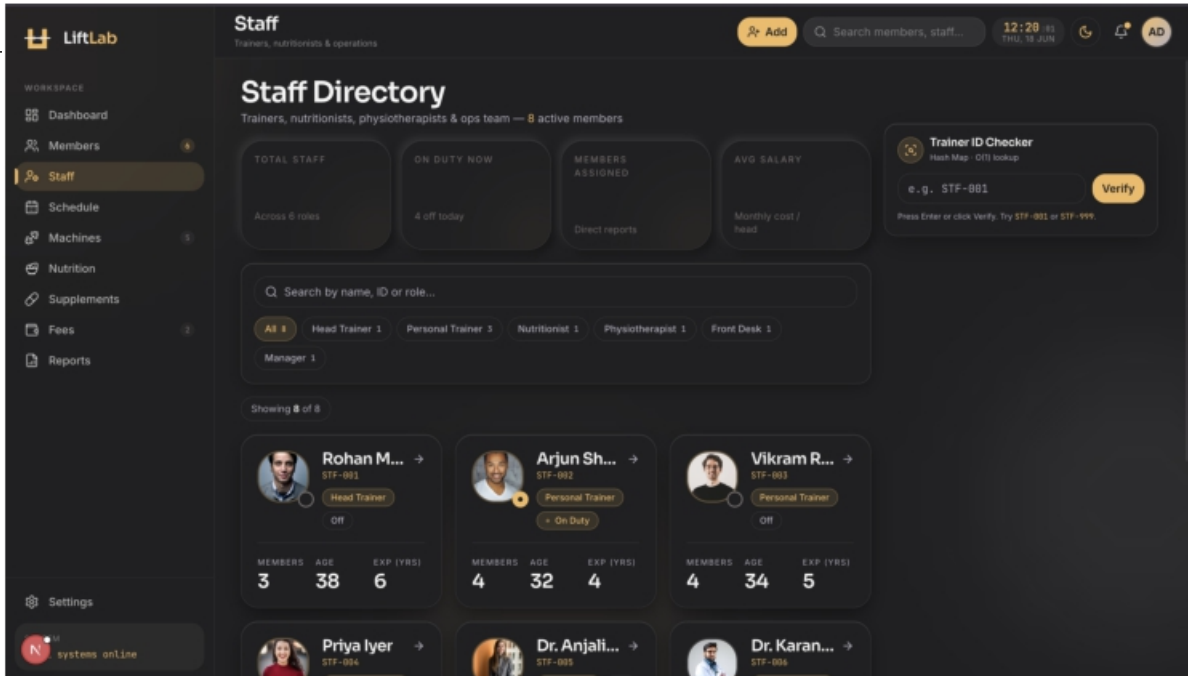


The **Add New Member** modal is displayed, allowing for the onboarding of a new member. The form includes the following sections:

- PERSONAL DETAILS:**
 - Full Name: e.g. Aryan Mehta
 - Email: member@email.com
 - Contact: +91-9876543210
 - Date of Birth: dd/mm/yyyy
 - Gender: Male
 - Blood Group: O+
 - Height (cm): 175
 - Weight (kg): 75
- MEMBERSHIP:**
 - Membership Type: Pro
 - Assigned Trainer: Rohan Mehta - Head Trainer
- EMERGENCY CONTACT:**
 - Contact Name: [Blank]
 - Contact Phone: [Blank]

The background shows a list of existing members with columns for ID, Member Name, Progress, Attendance, and Fees.

LiftLab - Gym Management Dashboard



Staff
Trainers, nutritionists & ops team — 8 active members

Staff Directory
Trainers, nutritionists, physiotherapists & ops team — 8 active members

Summary Cards:

- TOTAL STAFF:** Across 6 roles
- ON DUTY NOW:** 4 off today
- MEMBERS ASSIGNED:** Direct reports
- AVG SALARY:** Monthly cost / head

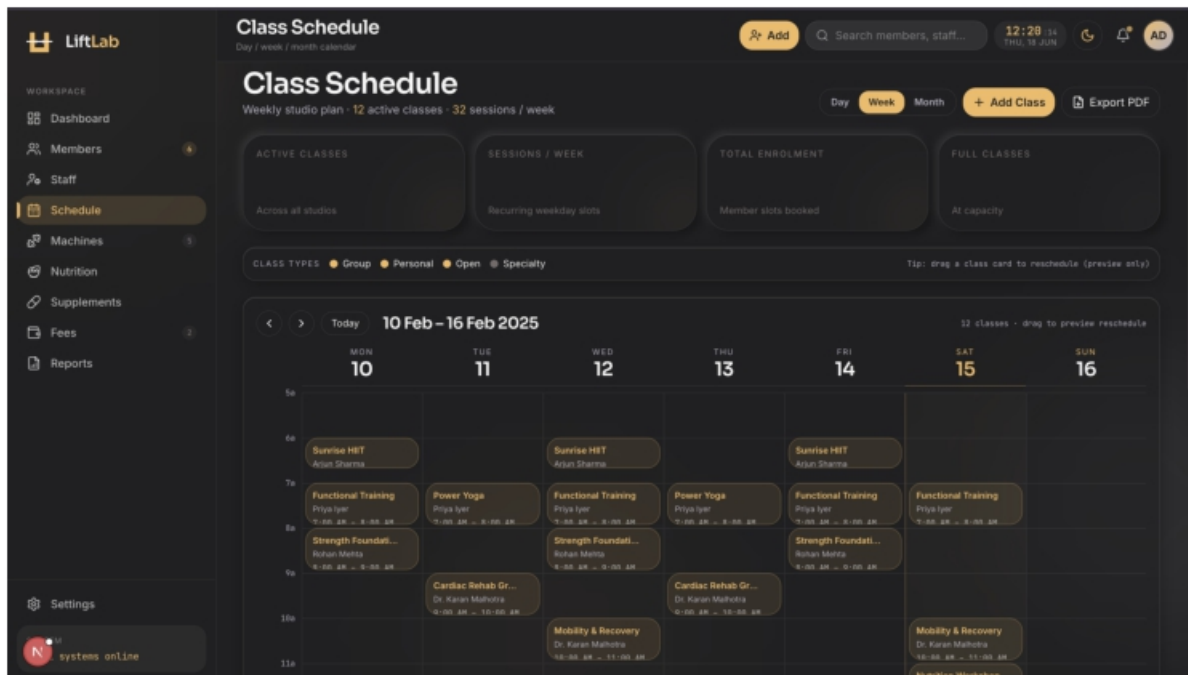
Trainer ID Checker
Hash Map - O(1) lookup
e.g. STF-001 **Verify**
Press Enter or click Verify. Try STF-001 or STF-999.

Search: Search by name, ID or role...

Filters: All 8 | Head Trainer 1 | Personal Trainer 3 | Nutritionist 1 | Physiotherapist 1 | Front Desk 1 | Manager 1

Showing 8 of 8

Staff Member	Role	Status	Members	Age	Exp (Yrs)
Rohan M...	Head Trainer	Off	3	38	6
Arjun Sh...	Personal Trainer	On Duty	4	32	4
Vikram R...	Personal Trainer	Off	4	34	5
Priya Iyer					
Dr. Anjali...					
Dr. Karan...					



Class Schedule
Day / week / month calendar

Class Schedule
Weekly studio plan · 12 active classes · 32 sessions / week

Summary Cards:

- ACTIVE CLASSES:** Across all studios
- SESSIONS / WEEK:** Recurring weekday slots
- TOTAL ENROLMENT:** Member slots booked
- FULL CLASSES:** At capacity

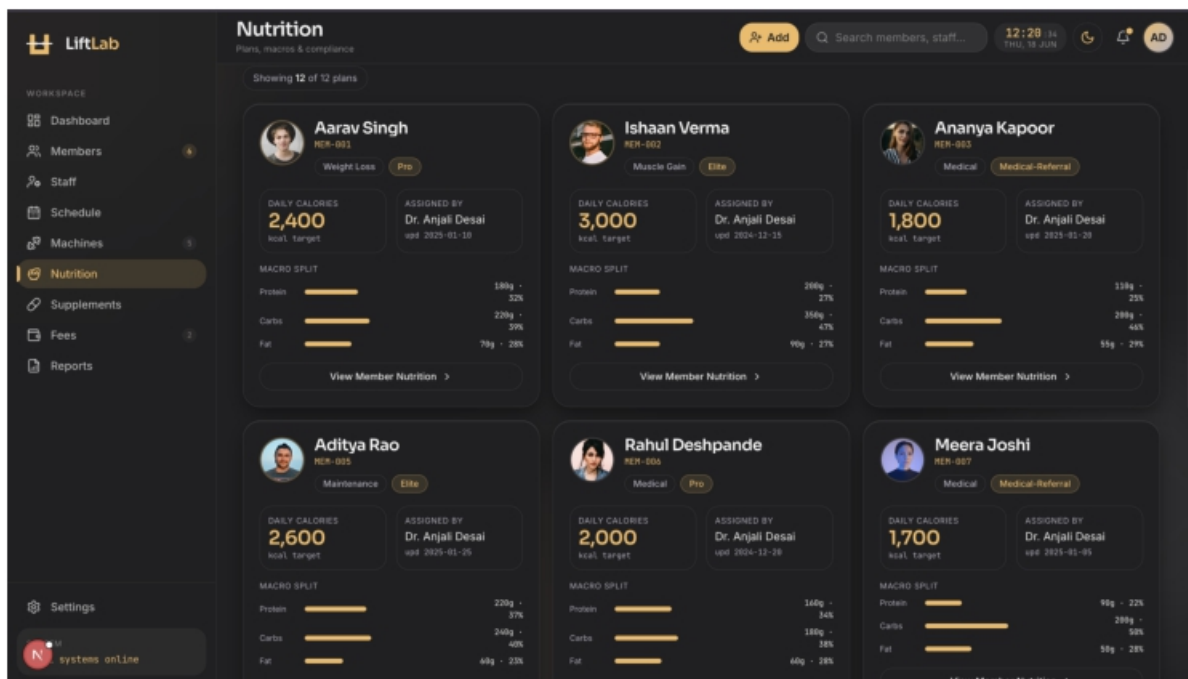
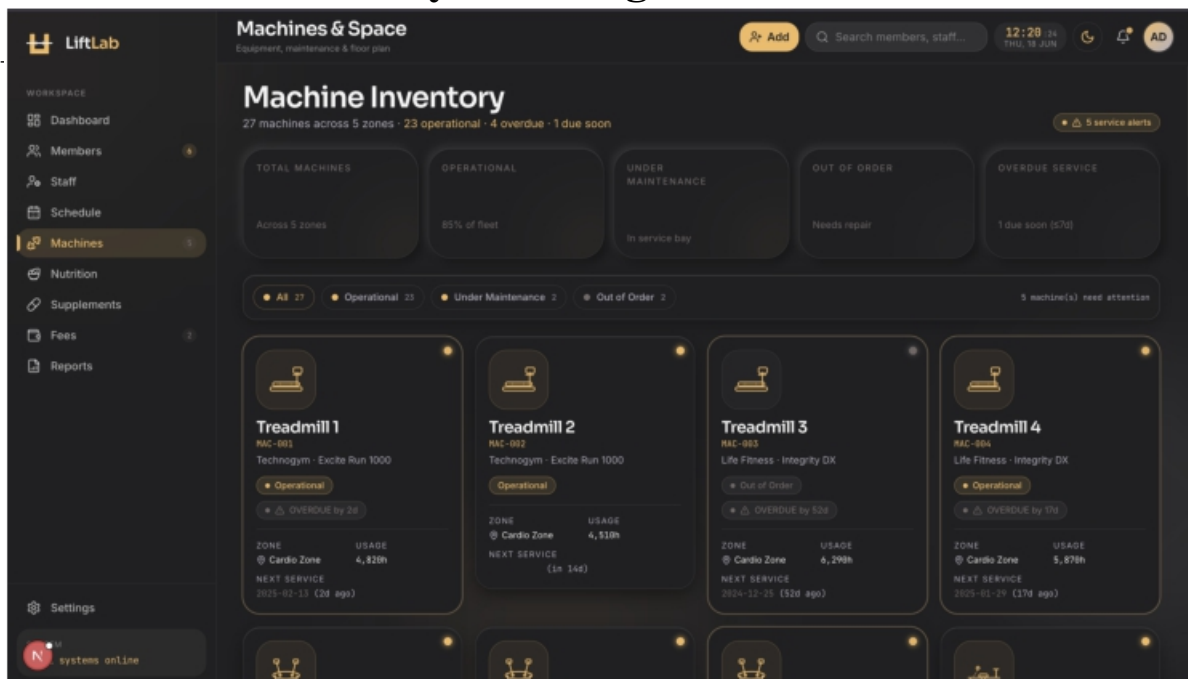
CLASS TYPES: Group | Personal | Open | Specialty

Tip: drag a class card to reschedule (preview only)

Calendar View: 10 Feb - 16 Feb 2025

	MON 10	TUE 11	WED 12	THU 13	FRI 14	SAT 15	SUN 16
5a							
6a	Sunrise HIT Arjun Sharma		Sunrise HIT Arjun Sharma		Sunrise HIT Arjun Sharma		
7a	Functional Training Priya Iyer 9:00 AM - 9:55 AM	Power Yoga Priya Iyer 9:00 AM - 9:55 AM	Functional Training Priya Iyer 9:00 AM - 9:55 AM	Power Yoga Priya Iyer 9:00 AM - 9:55 AM	Functional Training Priya Iyer 9:00 AM - 9:55 AM	Functional Training Priya Iyer 9:00 AM - 9:55 AM	
8a	Strength Foundati... Rohan Mehra 9:00 AM - 9:55 AM		Strength Foundati... Rohan Mehra 9:00 AM - 9:55 AM		Strength Foundati... Rohan Mehra 9:00 AM - 9:55 AM		
9a		Cardiac Rehab Gr... Dr. Karan Mahotra 9:00 AM - 10:00 AM		Cardiac Rehab Gr... Dr. Karan Mahotra 9:00 AM - 10:00 AM			
10a			Mobility & Recovery Dr. Karan Mahotra 10:00 AM - 11:00 AM			Mobility & Recovery Dr. Karan Mahotra 10:00 AM - 11:00 AM	
11a						Nutrition Workshop	

LiftLab - Gym Management Dashboard



LiftLab - Gym Management Dashboard



WORKSPACE

- Dashboard
- Members
- Staff
- Schedule
- Machines
- Nutrition
- Supplements**
- Fees
- Reports

systems online

Supplements & Products

History, sponsors & sales

Add

Search members, staff...

12:28 THU, 18 JUN

AD

Supplements & Merch

18 products across 8 categories · 6 brand partners · 4 low-stock

4 restock alerts

TOTAL PRODUCTS

6 brand partners

IN STOCK

78% of catalog

LOW STOCK

< 5 units left

OUT OF STOCK

Needs reorder

AVG MEMBER OFF

Across all products

Sponsor Management

6 active brand partnerships · 0 expiring within 90d

BRAND	DEAL TYPE	DISCOUNT	EXPIRY	CONTACT
MuscleBlaze	Official Partner	26% off		partnerships@muscleblaze.com
Optimum Nutrition	Retail Stockist	15% off		india@optimumnutrition.com
Myprotein	Endorsement	25% off		india-82@myprotein.com
GNC	Retail Stockist	10% off		partners@gncindia.in
AS-IT-IS	Athlete Sponsor	18% off		athletes@asitision.com
Ultimate Nutrition	Endorsement	12% off		india@ultimagenutrition.com

All 18

Whey Protein

Creatine

BCAA

Pre-Workout

Fat Burner

Vitamins

Accessories

Apparel



WORKSPACE

- Dashboard
- Members
- Staff
- Schedule
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- Nutrition
- Supplements
- Fees**
- Reports

systems online

Fees & Finance

Billing lifecycle & revenue

Add

Search members, staff...

12:21 THU, 18 JUN

AD

MEMBER	PLAN	DUE DATE	PAID	BALANCE	STATUS	LAST PAYMENT
Aarav Singh M01-001	Pro	2025-04-15	₹22,500	₹2,500	Partial	2025-01-15 UPI - RC-2431-023
Ishaan Verma M01-002	Elite	2025-06-01	₹48,000	₹0	Paid	2024-06-01 Bank Transfer - RC-2406-001
Ananya Kapoor M01-003	Medical-Referral	2025-02-10	₹38,500	₹3,500	Paid	2024-11-10 Bank Transfer - RC-2411-122
Kabir Khanna M01-004	Basic	2025-07-01	₹6,000	₹3,000	Partial	2024-12-15 UPI - RC-2412-088
Aditya Rao M01-005	Elite	2025-03-20	₹48,000	₹0	Paid	2024-03-20 Bank Transfer - RC-2403-020
Rahul Deshpande M01-006	Pro	2025-01-15	₹20,000	₹7,500	Overdue	2024-01-15 UPI - RC-2401-023
Meera Joshi M01-007	Medical-Referral	2025-09-15	₹42,000	₹0	Paid	2024-06-15 UPI - RC-2406-110
Sahil Malhotra M01-008	Pro	2024-12-20	₹20,000	₹10,000	Partial	2024-06-20 Bank Transfer - RC-2406-110
Vivaan Gupta M01-009	Elite	2025-09-10	₹48,000	₹0	Paid	2024-09-10 Bank Transfer - RC-2409-010
Diya Sharma M01-010	Basic	2025-05-01	₹9,000	₹0	Paid	2024-11-01 Cash - RC-2411-088
Arnav Pillai M01-011	Pro	2025-01-05	₹15,000	₹12,500	Overdue	2024-07-05 Cash - RC-2407-088
Kavya Reddy M01-012	Elite	2025-06-10	₹48,000	₹0	Paid	2024-06-10 Bank Transfer - RC-2406-010

LiftLab - Gym Management Dashboard

6. DECLARATION

I hereby declare that this project work entitled **Design and Development of LIFTLAB - Gym Management Dashboard Using Next.js, React, and JavaScript** is my original work carried out by me. This work has not been copied from any source and has not been submitted earlier for any examination.

Name of Student: Raj Bharat Rasal

Roll Number : 150096725066

Signature of Student:



Date: 18 June 2026

LiftLab - Gym Management Dashboard

7. FUTURE SCOPE

The current LiftLab dashboard is built with Next.js, React, and client-side Zustand stores. The future enhancements that can be implemented are:

- Adding a real backend API with Prisma and SQLite for persistent member, staff, and transaction data storage.
- Implementing user authentication with NextAuth.js for role-based access (admin, trainer, nutritionist, front desk).
- Adding real-time WebSocket updates for live sensor data instead of simulated intervals.
- Integrating payment gateway (Razorpay/Stripe) for online fee collection and automatic receipt generation.
- Building a mobile companion app using React Native for member check-ins and workout tracking.
- Adding AI-powered workout plan recommendations based on member progress data and goals.
- Implementing email and SMS notifications for fee reminders, class schedules, and membership expiry alerts.
- Enhancing accessibility with full WCAG 2.1 AA compliance, keyboard navigation, and screen reader optimization.
- Deploying to production with Vercel or a custom server, with CDN for static assets and image optimization.